

Concept Relationship Embedding-Based Interactive Web Application for Explainable Medical Diagnosis

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Abstract

Deep learning has made remarkable progress in medical image analysis, yet its black-box nature still limits interpretability and clinician trust. Concept-based modeling offers a promising direction for explainable AI by integrating human-understandable concepts. However, existing approaches typically rely on global concept annotations and infer diagnosis based solely on the presence or absence of individual concepts. This oversimplified paradigm ignores the rich relationships among concepts and their causal influence on disease outcomes. To overcome these limitations, we propose the Concept Relationship Embedding Model (CREM) for interpretable medical diagnosis. CREM mirrors coarse-to-fine clinical reasoning by first extracting fine-grained subregional concepts, then explicitly encoding their relationships as a concept interaction graph, and finally performing causal inference between concepts and diagnoses to enable reliable and transparent diagnostic predictions. We evaluate CREM on four public medical imaging benchmarks, where it achieves state-of-the-art performance on both concept recognition and disease classification tasks, while exhibiting improved robustness, label efficiency, and interpretability. Furthermore, we deploy CREM as an interactive web-based demo that allows clinicians to visualize concept activations, trace diagnostic reasoning paths, and iteratively refine concept cues, facilitating effective human-in-the-loop decision-making.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Life and medical sciences**.

Keywords

Explainable Artificial Intelligence, Concept-Based Models, Interactive Diagnosis Copilot

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1 Introduction

Black-box deep learning methods have emerged as powerful tools in medical image analysis [7, 42, 58, 59], with significant potential to revolutionize healthcare diagnosis and treatment, such as skin cancer detection [16, 19] and thyroid nodule risk stratification [14, 50]. Despite their promising performance, the end-to-end opaque nature of black-box deep learning methods has sparked a crisis of trust regarding the interpretability and reliability of model predictions, particularly in fields requiring high accountability, such as healthcare [35, 41, 44, 56]. Decisions made by deep learning models in high-risk clinical settings are directly related to human life and health [18, 55]. Therefore, model decisions must be adequately explained to medical experts and patients to promote trust in model predictions, enable informed clinical decision-making, and allow AI systems to further enhance human well-being [39, 57]. This has spurred a growing focus on research on Explainable Artificial Intelligence (XAI), highlighting that explainability is essential to foster trust, improve clinical decision-making, and ultimately enhance patient safety and quality of care [6, 34, 40].

The main XAI methods are broadly categorized into post-hoc and ante-hoc approaches [2, 12]. Post-hoc methods, such as saliency maps [31, 45, 48] and gradient-based techniques [5, 46], interpret model decisions after training by highlighting key image regions or features influencing predictions. However, recent studies [3, 25, 49] have shown that these methods are often subjective and sensitive to slight variations in input data, which can misrepresent the model's underlying decision-making process. To overcome these limitations, researchers have shifted focus toward ante-hoc XAI methods, which embed interpretability directly into the model's architecture or learning process, offering clearer and more reliable insights into how decisions are made.

Ante-hoc XAI methods primarily include attention-based models [29, 30], example-based models [17, 24], and concept-based



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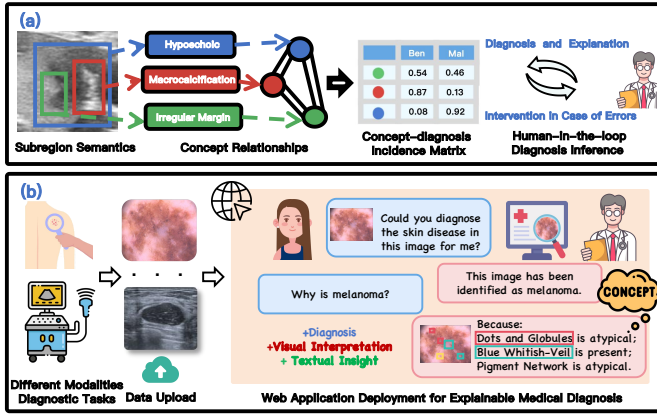


Figure 1: Our method mimics the clinical diagnostic workflow by performing causal inference based on concepts learned through graph reasoning. The interpretable diagnostic model is deployed as a user-friendly interactive web application to support clinical decision-making.

models [6, 36]. Recently, concept-based models have emerged as a promising ante-hoc framework that leverages human-interpretable concepts as intermediate representations, offering a unique paradigm of interpretability, intervenability, and predictability. Concept-based models first map input features to a set of high-level human-understandable concepts and use these concepts to solve downstream tasks. As shown in Figure 1(a), in medical diagnosis, models do not directly derive diagnoses from raw images (**image** $\Rightarrow$ **diagnosis**). Instead, concepts are defined as high-level attributes of lesions or abnormalities and serve as evidence for diagnostic outcomes (**image** $\Rightarrow$ **concept** $\Rightarrow$ **diagnosis**).

Compared to black-box methods, existing concept-based models face a major challenge: model performance is often compromised owing to architectural constraints of concept bottlenecks, particularly when concept labels are weakly informative. For example, the concept embedding model [13] mitigates this trade-off by employing high-dimensional concept representations with both positive and negative semantics within the bottleneck layer. Similarly, the post-hoc concept bottleneck model [53] incorporates a residual modeling step to bridge the performance gap between concept learning and the original black-box model. Despite recent advances, existing concept-based models for explainable medical diagnosis still suffer from the following key limitations:

(1) **Concept Embedding Inefficiency.** A medical image usually contains multiple concepts, such as different types of lesions or abnormalities in different regions. Existing methods utilize concept labels to supervise concept predictions at the image level, neglecting fine-grained semantic relationships between subregions and their corresponding concepts.

(2) **Unconsidered Concept Relationships.** Co-varying concepts provide critical contextual cues for human understanding, particularly in healthcare, where co-occurring symptoms inform diagnosis. However, existing models focus solely on the presence or absence of individual concepts, neglecting their interactions in diagnostic reasoning.

In response to these limitations, we propose the Concept Relationship Embedding Model (CREM), which integrates concept-based modeling with relational graph reasoning, as illustrated in Figure 1. CREM simulates a coarse-to-fine clinical diagnosis process: it first extracts features from localized subregions of a medical image (coarse stage), then models the relationships among these features to facilitate information and contextual sharing (fine stage), and finally generates an overall diagnostic conclusion by integrating the activated diagnostic concepts (diagnostic stage). In addition, we develop an interactive user-friendly web demo that operationalizes CREM for an explainable medical diagnosis. The platform enables clinicians to visualize concept activations, trace diagnostic reasoning pathways, and interactively adjust concept cues, supporting human-in-the-loop decision-making. Our contributions are summarized as follows:

(1) The Concept Relationship Reasoning Bottleneck (CRRB) is proposed to incorporate graph reasoning with coarse-grained relationships and fine-grained semantic knowledge, overcoming the limitations of existing concept-based models in relational reasoning for disease diagnosis.

(2) The Concept-Diagnosis Causal Inference (CDCI) module is designed to effectively leverage biases between concept-aware and diagnosis-prediction spaces, surpassing the performance of black-box models.

(3) Benefiting from efficiently extracting concept-aware sub-region embeddings and modeling holistic interactions between concepts and diseases, CREM is robust enough to withstand challenging settings, including limited training data regimes, sparse concept supervision, and real-world concept distortions.

(4) This study establishes an intelligent framework that integrates CREM with an interactive web demo, bridging the gap between concept-based reasoning and disease diagnosis while providing an intuitive and efficient solution for explainable medical diagnosis. Extensive experiments on four public benchmarks demonstrate that CREM delivers visual insights and textual explanations, enhancing performance, preserving interpretability, and enabling human-in-the-loop intervention.

2 Related Work

Incorporating human-understandable units, called Concepts, into deep learning has been seen as a promising approach to improve model explainability. Recently, concept-based explainable models have been widely studied. Concept Activation Vectors (CAVs) [15, 21] train linear classifiers on feature spaces to verify whether representations can be reduced to human-defined conceptual examples. Concept Bottleneck Models (CBMs) [27, 47, 54] first predict concepts, and then use the detected concepts to predict task labels. CAVs require additional probe datasets, while CBMs describe concepts as independent values annotated by experts. Some researchers also explore automatic concept discovery [28, 52], but these methods may not be suitable for healthcare due to potential mismatches with established medical terminology, which can compromise diagnostic accuracy. Although some concept-based methods focus on improving task performance or reducing annotation effort [37, 38], they solely determine the presence of concepts, ignoring the complex dependencies and structures among them.

To address this issue, our study mimics a coarse-to-fine diagnostic process through graph reasoning and causal inference to improve user confidence in computer-aided diagnostics.

3 Method

3.1 Overall Framework

Figure 2 illustrates the overall architecture of the Concept Relationship Embedding Model (CREM) for explainable disease diagnosis. (1) The concept-aware encoder uses a Transformer-based architecture guided by concept queries to adaptively focus on subregions, injecting category-specific semantics and spatial details into concept-aware embeddings. (2) These concept-aware embeddings are then processed by the Concept Relationship Reasoning Bottleneck (CRRB), which uses a graph attention mechanism to learn coarse- and fine-grained concept relationships, aggregating diagnosis-critical concept embeddings with minimal redundancy. (3) To model the causal influence of concepts on disease diagnosis, the Concept-Diagnosis Causal Inference (CDCI) module learns a concept-diagnosis incidence matrix, explicitly quantifying the correlations between concept sets and diagnosis outcomes. (4) CREM ensures interpretability, reliability, and intervenability by unifying concept detection and disease classification within an end-to-end framework.

3.2 Concept-aware Encoder

Different diseases or abnormalities often manifest as distinct localized regions in medical image analysis, such as irregular boundaries or localized ulcerations in skin disease detection. To achieve accurate diagnosis, it is essential to adaptively extract distinct subregion features corresponding to specific concept categories. Inspired by prompt learning [32], the concept-aware encoder employs multi-head attention to selectively attend to target subregions and generate concept-specific sequences under the guidance of prompt queries.

Given an image $X \in \mathbb{R}^{H \times W \times 3}$ as input, the image feature encoder first extracts spatial visual features $I_x \in \mathbb{R}^{H_1 \times W_1 \times D_1}$. A linear projection layer is then applied to project the features into $F_x \in \mathbb{R}^{H_1 \times W_1 \times D}$, ensuring dimensional alignment with subsequent query-based attention operations. The concept-aware encoder adopts a Transformer-based architecture composed of self-attention, cross-attention, and feed-forward network (FFN) modules. Specifically, a set of K concept categories $C = \{c_1, \dots, c_K\}$ is encoded by a text encoder and used to initialize concept queries $C_0 = \{c_{(0,1)}, \dots, c_{(0,K)}\} \in \mathbb{R}^{K \times D}$, where each query $c_{(0,k)} \in \mathbb{R}^{1 \times D}$ corresponds to a specific concept category. These concept queries are treated as learnable parameters and are updated in parallel across N Transformer decoder layers. At the n -th layer, the query updating process is defined as:

$$\begin{aligned} C_n^{(1)} &= \text{SelfAttn}(\hat{C}_{n-1}, \hat{C}_{n-1}, \hat{C}_{n-1}), \\ C_n^{(2)} &= \text{CrossAttn}(\hat{C}_n^{(1)}, \hat{F}_x, \hat{F}_x), \\ C_n &= \text{FFN}(C_n^{(2)}), \end{aligned} \quad (1)$$

where $\text{SelfAttn}(\cdot)$ and $\text{CrossAttn}(\cdot)$ denote standard multi-head attention operations. In the self-attention module, query, key and value are all from concept embeddings, whereas in the cross-attention module, key and value are drawn from the visual features. The hat

notation indicates that sinusoidal positional encodings [4] are added to incorporate spatial context. The cross-attention mechanism iteratively refines concept embeddings by attending to relevant visual features and aggregating informative cues. Through layer-wise updates, the concept embeddings progressively incorporate contextual information from the input image. The concept-aware encoder outputs the final concept representations $C_N \in \mathbb{R}^{K \times D}$, where each concept is treated as a binary classification task to predict concept probabilities $p_c = \text{sigmoid}(\text{Linear}(C_N))$.

3.3 Concept Relationship Reasoning Bottleneck

Previous concept-based bottlenecks map input data into isolated concept embeddings or scalar concept scores for downstream diagnostic tasks. However, by failing to capture higher-order and non-linear interactions between concepts, they often result in degraded prediction accuracy and a trade-off between performance and interpretability. To address these challenges, we design a Concept Relationship Reasoning Bottleneck (CRRB) that mimics the clinical diagnostic process. CRRB effectively captures relationships between concepts through coarse- and fine-grained graph reasoning, while preserving the embeddings most critical for the diagnostic task.

Coarse-grained relationship learning: Based on the concept-aware embeddings $C_N = \{c_{(N,1)}, \dots, c_{(N,K)}\}^T$ generated by the concept-aware encoder, the coarse-grained relationship learning module constructs an adaptive concept-to-concept undirected graph $G_{\text{coarse}} = \langle C, A \rangle$, where each node corresponds to a specific concept. The adjacency matrix $A \in \mathbb{R}^{K \times K}$ is defined in the latent concept space as:

$$A_{(i,j)} = \frac{c_{(N,i)} \cdot c_{(N,j)}}{\|c_{(N,i)}\| \cdot \|c_{(N,j)}\|}, \quad (2)$$

where i and j index the K concept nodes. The adjacency matrix A encodes the connectivity by measuring the similarity between concept embeddings. Next, a graph convolutional layer is applied to extract topological information for each node in the concept graph. The output evaluates the importance of each concept node based on its structural information:

$$S_{\text{coarse}} = \sigma \left(D^{-\frac{1}{2}} A D^{-\frac{1}{2}} C_N W \right), \quad (3)$$

where S_{coarse} denotes the structural information score of each node, D is the degree matrix of the adjacency matrix A , W is a trainable weight matrix, and $\sigma(\cdot)$ denotes the sigmoid activation function.

Fine-grained semantic learning: Each node in the concept graph contains specific semantic information. The fine-grained semantic learning module explicitly determines whether each concept node is in an active state for the downstream diagnostic task. We adopt a single-layer MLP as a learnable and differentiable scoring function S_{fine} , which evaluates the activity degree of each concept within the concept-aware embedding space. The final concept activity score is computed as:

$$S_{\text{fine}} = \sigma(\text{MLP}(C_N)), \quad (4)$$

where C_N denotes the node embedding matrix and $\sigma(\cdot)$ represents the sigmoid activation function. S_{fine} indicates the concept activation probability. Finally, the importance of each node is determined by ranking their scores $S = S_{\text{coarse}} + S_{\text{fine}}$.

Diagnosis-critical embedding aggregation: On the one hand, not all concept-aware embeddings contribute equally to disease

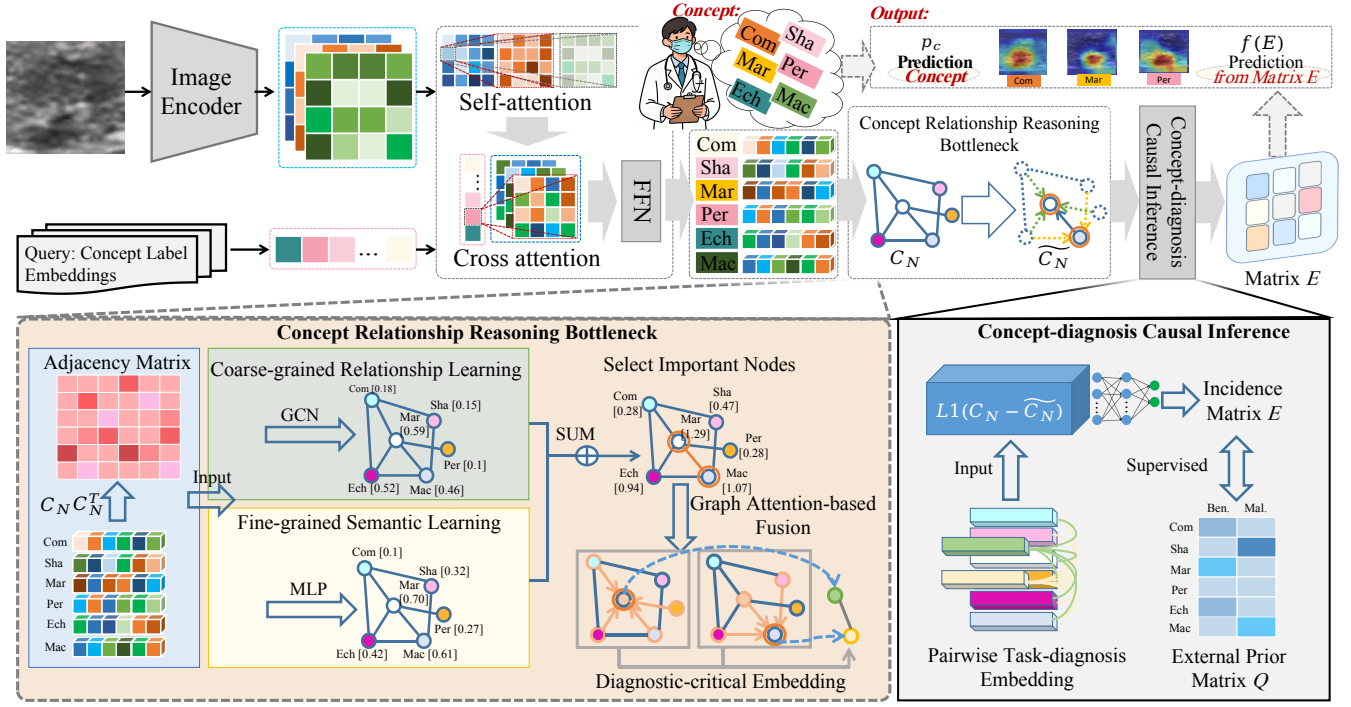


Figure 2: Overall pipeline of the proposed framework. The concept-aware encoder extracts category-specific subregion features to form concept embeddings. These embeddings are processed by a Concept Relationship Reasoning Bottleneck (CRRB), which models inter-concept dependencies through coarse graph learning and fine semantic scoring, and aggregates diagnosis-critical concept embeddings via graph attention. The Concept-Diagnosis Causal Inference (CDCI) module further learns a concept-disease incidence matrix to quantify causal contributions to diagnostic outcomes, enabling end-to-end interpretable and reliable disease prediction.

prediction. Feeding all node embeddings into downstream diagnostic predictors can lead to feature redundancy. On the other hand, some concepts are more predictive of disease diagnosis than others. Therefore, more attention should be given to nodes corresponding to more indicative embeddings in the concept graph. We select the top T concept nodes with the highest importance scores, where T equals the number of diagnostic categories.

To incorporate information from non-selected concept nodes, we propose a graph attention-based fusion method that aggregates embeddings from adjacent non-selected nodes into the selected ones. The graph attention mechanism computes attention scores over neighboring nodes and uses them as weights to update the embeddings of selected nodes:

$$\tilde{c}_{(N,i)} = \sigma \left(\sum_{j \in N_i} \alpha_{(i,j)} \tilde{w} c_{(N,j)} \right), \quad (5)$$

$$\alpha_{(i,j)} = \frac{\exp \left(\text{LR} \left(a^T [\tilde{w} c_{(N,i)} \parallel \tilde{w} c_{(N,j)}] \right) \right)}{\sum_{m \in N_i} \exp \left(\text{LR} \left(a^T [\tilde{w} c_{(N,i)} \parallel \tilde{w} c_{(N,m)}] \right) \right)}, \quad (6)$$

where $c_{(N,i)}$ denotes the embedding of concept node i , N_i denotes the set of neighboring nodes of node i , $\text{LR}(\cdot)$ denotes the LeakyReLU activation function, and $\alpha_{(i,j)}$ is the attention coefficient between

nodes i and j . In this manner, the aggregated concept representations $\tilde{C}_N \in \mathbb{R}^{T \times D}$ encode informative graph representations that highlight diagnosis-critical concept sets. The resulting embeddings are then passed through a linear layer for final disease classification: $p_y = \text{softmax}(\text{Linear}(\tilde{C}_N))$.

3.4 Concept-Diagnosis Causal Inference

To better capture how characteristics of the concept set influence diagnosis performance, we design a **Concept-Diagnosis Causal Inference (CDCI) module** to model the causal relationships between concepts and the final disease decision. Specifically, we introduce a **concept-diagnosis incidence matrix** $E \in \mathbb{R}^{K \times T}$, which is learned to capture inductive associations between the concept and disease spaces. Each element $e_{(i,j)}$ in E models the pairwise relationship between concept i and disease j , and is learned through a stacked MLP:

$$e_{(i,j)} = \text{MLP}_E(\ell_1(c_{(n,i)}, \tilde{c}_{(n,j)})), \quad (7)$$

where the ℓ_1 difference is computed between the concept-aware embedding $c_{(n,i)}$ and the disease prediction embedding $\tilde{c}_{(n,j)}$.

Based on ground-truth concept and disease labels, we construct a prior ground-truth matrix $Q \in [0, 1]^{K \times T}$ for each image, where each entry $q_{(i,j)}$ indicates whether concept i has a causal contribution to diagnosing disease j . For example, $q_{(i,j)} = 1$ indicates a strong

causal relationship, while $q_{(i,j)} = 0$ suggests no direct influence. The MLP is trained to align $e_{(i,j)}$ with $q_{(i,j)}$ by minimizing a consistency loss:

$$L(E, Q) = \sum_{i=1}^K \sum_{j=1}^T (e_{(i,j)} - q_{(i,j)})^2. \quad (8)$$

This explicit supervision encourages the model to learn meaningful causal connections through the learned matrix E , facilitating accurate disease prediction and enhancing interpretability by highlighting the contribution of each concept to the diagnostic process. The matrix E is then applied through a linear predictor $f(E)$ to map concept–diagnosis causal relationships to final disease predictions. The learned weights in E reflect the importance of each concept for diagnosis, enabling human experts to adjust the matrix to obtain more reliable diagnostic results when errors are observed.

4 Experiments

4.1 Experimental Setup

Datasets. *Derm7pt* [20] comprises 1,011 dermoscopic images, with 2 diseases (nevus and melanoma) and 12 concept groups from the 7-Point checklist. *SkinCon* [9], the most comprehensive dermatology dataset, includes 3,230 images annotated with 22 selected concept groups and categorized into malignant, benign, and non-neoplastic cases. *PBC* [1], a comprehensive resource for identifying blood-related conditions, contains 10,298 images annotated with 11 morphological attribute groups and categorized into five cell types: neutrophils, eosinophils, basophils, monocytes, and lymphocytes. *Thyroid* [50] consists of 12,855 ultrasound images annotated with 7 TI-RADS concept groups, classified as benign or malignant, supported by biopsy-confirmed pathological results. More details can be found in Table 5 of the Appendix.

Implementation Details. The training process is optimized with the Adam optimizer for 80 epochs, with a learning rate of 1×10^{-4} , a weight decay of 1×10^{-2} , and momentum parameters $(\beta_1, \beta_2) = (0.9, 0.999)$. For regularization, we apply Cutout [11] with a factor of 0.5, True Weight Decay [33] with a coefficient of 1×10^{-2} , and RandAugment [8] for image augmentation. Both our method and the benchmark models are implemented in PyTorch. CREM uses an ImageNet [10] pre-trained model as the image encoder and a frozen pre-trained CLIP text encoder [43] as the text encoder. CREM-b is the black-box variant that directly outputs diagnostic results.

Baseline and Evaluation Metrics. We compare CREM with state-of-the-art methods, including Concept Bottleneck Model (CBM) [27], Concept Embedding Model (CEM) [13], Probabilistic Concept Bottleneck Model (ProbCBM) [23], Energy-based Concept Bottleneck Model (ECBM) [51], and Concept Graph Embedding Model (CGEM) [26]. Experimental analysis focuses primarily on the precision of disease diagnosis and concept detection, as well as interpretability, intervention, label efficiency, and robustness. The evaluation metrics encompass the accuracy (ACC) and Area Under the Curve (AUC) for each individual concept, mean ACC_C and AUC_C across all concepts, and $F1_D$, ACC_D and AUC_D for disease diagnosis.

4.2 Accuracy Analysis

Disease Diagnosis. Table 1 shows that CREM outperforms existing concept-based methods while remaining competitive with the black-box model, without sacrificing interpretability. CREM achieves accuracy of 87.19%, 83.64%, 99.74%, and 88.88% and AUC of 92.23%, 88.93%, 100%, and 95.04% on the Derm7pt, SkinCon, PBC, and Thyroid datasets, respectively. Unlike previous methods that rely solely on the presence of individual concepts, CREM mimics a coarse-to-fine clinical diagnostic process through structured inter-concept reasoning and concept–diagnosis causal inference, thereby alleviating the traditional accuracy–interpretability trade-off.

Concept Detection. As shown in Table 1, all baselines achieve similar ACC_C but display poor AUC_C values. This discrepancy indicates that these approaches are biased toward predicting majority concepts and lead to poor performance in detecting minority concepts. In contrast, CREM effectively addresses concept imbalance, achieving better sensitivity across both majority and minority concept groups. CREM excels at identifying fine-grained concepts, as demonstrated in Figure 3, particularly for skin lesion patterns and subtle ultrasound tissue characteristics in medical images. However, other advanced methods struggle with challenging datasets, such as SkinCon and Thyroid, primarily due to class imbalance and imaging irregularities. Benefiting from the powerful concept encoder, CREM successfully captures subtle semantic relationships between subregions and concepts in medical images, consistently demonstrating superior concept detection performance. We further demonstrate that the concept importance and concept–diagnosis associations learned by our model are more closely aligned with clinical importance (Appendix Figure 10).

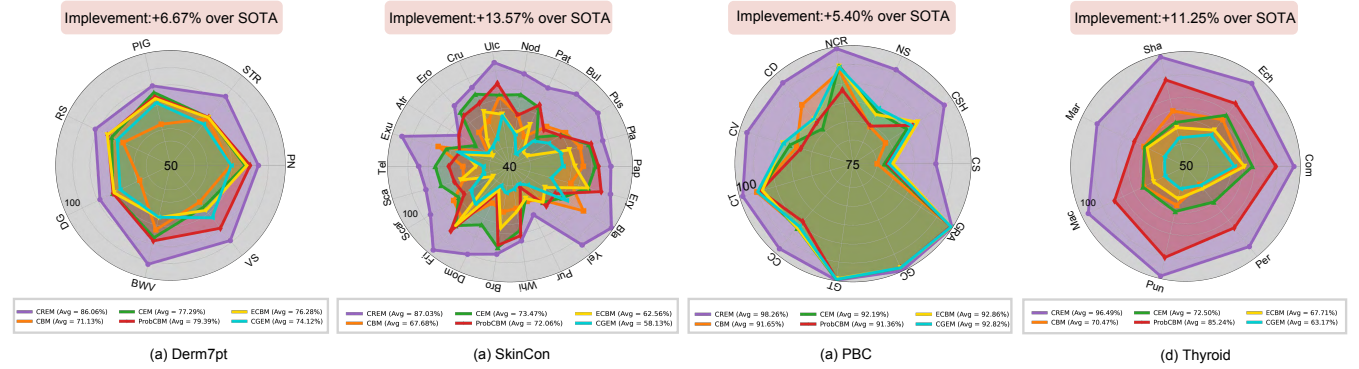
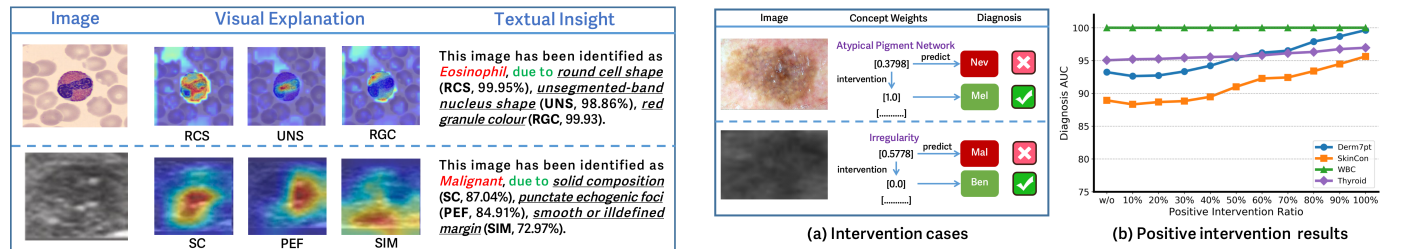
4.3 Explainability Analysis

Interpretability for Plausibility. CREM achieves both plausibility and interpretability by providing concept-based textual and visual explanations. As illustrated in Figure 4, given an input medical image, CREM produces a diagnosis along with the localization and contribution of each predicted concept. The textual explanation is derived from concepts that make positive contributions to the diagnostic decision, as indicated by the sigmoid outputs of the final linear layer. Meanwhile, the visual explanation highlights the image regions activated for each concept, thereby grounding the diagnostic reasoning in clinically meaningful evidence. By explicitly linking diagnoses to human-interpretable concepts and their associated regions, CREM facilitates an intuitive understanding of the decision-making process and enhances transparency and clinical reliability in real-world diagnostic scenarios.

Intervention for Faithfulness. Faithfulness requires that explanations faithfully reflect the underlying model decision mechanism. In CREM, we assess faithfulness through test-time concept intervention enabled by the learned concept–diagnosis incidence matrix, which quantifies concept importance and allows direct manipulation by human experts. During inference, we perform interventions on mispredicted concepts by adjusting the corresponding matrix weights and observe the resulting changes in diagnosis. As illustrated in Figure 5(a), correcting the weight of “atypical pigment

Table 1: Quantitative comparisons of disease diagnosis with state-of-the-art concept-based methods. The best and second-best results are highlighted in bold and underlined.

Dataset	Method	Disease Diagnosis			Concept Detection		Dataset	Method	Disease Diagnosis			Concept Detection	
		ACC_D	$F1_D$	AUC_D	ACC_C	AUC_C			ACC_D	$F1_D$	AUC_D	ACC_C	AUC_C
Derm7pt	CBM	82.19	80.30	81.92	74.88	71.13	SkinCon	CBM	78.05	63.95	72.48	81.20	67.68
	CEM	81.56	77.57	76.39	79.40	77.29		CEM	75.36	72.43	75.08	88.69	73.47
	ProbCBM	81.88	77.11	75.29	80.62	79.39		ProbCBM	70.39	63.88	76.25	91.19	72.06
	ECBM	86.07	85.81	86.19	80.27	76.28		ECBM	70.98	68.56	72.55	91.31	62.56
	CGEM	79.38	73.57	71.86	76.98	74.12		CGEM	72.67	66.41	67.42	89.48	53.55
	CREM	87.19	84.51	93.23	82.31	86.06		CREM	83.64	73.67	88.93	92.87	87.03
	CREM-b	83.44	78.85	92.83	N/A	N/A		CREM-b	82.82	71.87	89.14	N/A	N/A
	CBM	99.74	99.67	99.82	94.69	91.65	Thyroid	CBM	70.89	68.87	74.26	80.47	70.47
	CEM	99.65	99.52	99.69	95.62	92.19		CEM	70.99	69.43	74.64	79.45	72.50
	ProbCBM	99.26	98.95	99.40	95.23	91.36		ProbCBM	88.17	85.94	86.63	94.93	85.24
	ECBM	96.60	96.33	97.77	95.41	92.86		ECBM	83.91	83.75	84.48	82.56	67.71
	CGEM	99.65	99.47	99.68	95.75	92.82		CGEM	70.85	69.43	75.47	81.84	63.17
	CREM	99.74	99.67	100.00	95.94	98.26		CREM	88.88	86.80	95.04	95.40	96.49
	CREM-b	99.71	99.60	100.00	N/A	N/A		CREM-b	87.03	84.48	94.38	N/A	N/A
PBC	CBM	99.74	99.67	99.82	94.69	91.65	Thyroid	CBM	70.89	68.87	74.26	80.47	70.47
	CEM	99.65	99.52	99.69	95.62	92.19		CEM	70.99	69.43	74.64	79.45	72.50
	ProbCBM	99.26	98.95	99.40	95.23	91.36		ProbCBM	88.17	85.94	86.63	94.93	85.24
	ECBM	96.60	96.33	97.77	95.41	92.86		ECBM	83.91	83.75	84.48	82.56	67.71
	CGEM	99.65	99.47	99.68	95.75	92.82		CGEM	70.85	69.43	75.47	81.84	63.17
	CREM	99.74	99.67	100.00	95.94	98.26		CREM	88.88	86.80	95.04	95.40	96.49
	CREM-b	99.71	99.60	100.00	N/A	N/A		CREM-b	87.03	84.48	94.38	N/A	N/A
	CBM	99.74	99.67	99.82	94.69	91.65		CBM	70.89	68.87	74.26	80.47	70.47
	CEM	99.65	99.52	99.69	95.62	92.19		CEM	70.99	69.43	74.64	79.45	72.50
	ProbCBM	99.26	98.95	99.40	95.23	91.36		ProbCBM	88.17	85.94	86.63	94.93	85.24

**Figure 3: Fine-grained concept detection across various datasets, with each axis showing per-concept AUC (%).****Figure 4: Illustrating model interpretability for visual insight and textual explanation plausibility.****Figure 5: Model interventions for concept-based faithfulness.**

network” shifts the diagnosis from nevus to melanoma, while nullifying the weight of “irregularity” changes the prediction from malignant to benign, which is consistent with expert findings. Figure 5(b) further shows that task accuracy improves as more erroneous concepts are corrected through intervention across datasets. These

results demonstrate CREM’s capability to establish faithful causal relationships and collaborate effectively with experts to correct errors.

4.4 Label Efficiency Analysis

Sparse Concept Groups of Training Data. We compare the robustness of all baselines under sparse concept annotations by randomly subsampling concept groups from the dermatology datasets, which serve as supervision for all models during training. As observed in Figure 6, the task AUC of CREM is only slightly affected by the reduction in supervision. This demonstrates that CREM’s label efficiency is achieved by leveraging sparsely annotated data to establish strong correlations between concepts and diagnostic decisions within the causal inference module. As a result, CREM reduces the reliance on extensive concept annotations and their associated costs, which is particularly important in the medical domain.

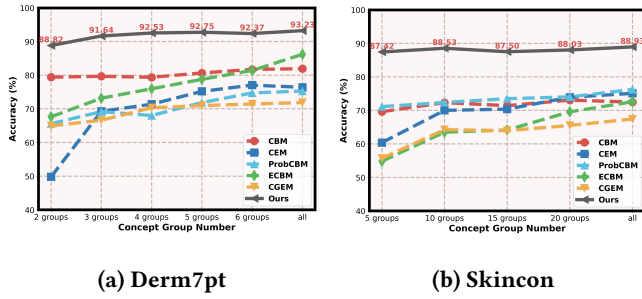


Figure 6: Performance on the diagnosis task with a subset of dermatology concepts during training.

Different Portion of Training Data. We evaluate the label efficiency of CREM by training with varying proportions of training data (10%-100%) while retaining the original full test data for testing. As shown in Figure 7, CREM achieves competitive performance in both concept detection and disease diagnosis on the SkinCon, PBC, and Thyroid datasets using only 10% of the training data. For the Derm7pt dataset, performance is suboptimal when only 10% of training data is used, likely due to the limited sample size (approximately 34 images). These results suggest that CREM achieves label efficiency by leveraging high-level relational abstractions instead of relying solely on large-scale data for feature extraction, highlighting its potential in label-limited scenarios.

Table 2: Comparison of diagnostic AUC (%) on the SkinCon dataset with uncertain concept annotations. w/o: without concept distortions; w: with distortions.

*: indicates performance parity; ↓: indicates performance decline.

Method	CBM	CEM	ProbCBM	ECBM	CGEM	CREM
w/o	72.48	75.08	76.25	72.55	67.42	88.93
w/	72.56 (*)	72.48 (↓)	68.30 (↓)	69.15 (↓)	67.10 (*)	88.58 (*)

4.5 Robustness Analysis

Most concept-based models rely solely on concept presence for decision-making, limiting their expressiveness and robustness to distortions. In dermatological diagnosis, we evaluate model robustness using pseudo-concept labels generated by a dermatosis foundation model [22], while keeping the original test set unchanged. Both embedding-based concept representations (CEM, CGEM) and scalar-based concept representations (CBM, ProbCBM, ECBM) exhibit varying degrees of performance degradation in downstream tasks due to the incompleteness or noisiness of pseudo-labels, as shown in Table 2. Our approach leverages graph-based reasoning to enhance expressiveness and robustness, enabling CREM to mitigate errors caused by concept distortions through relational learning.

Table 3: Ablation study of disease diagnosis on the Derm7pt dataset. The best results are highlighted in bold font.

CRRB	CDCI	ACC_D	$F1_D$	AUC_D
✗	✗	85.62	82.46	80.96
✓	✗	85.31	82.13	91.27
✗	✓	86.56	84.05	91.21
✓	✓	87.19	84.51	93.23

Table 4: Ablation study of CREM backbone.

Backbone	Derm7pt		Skincon	
	AUC_D	AUC_C	AUC_D	AUC_C
CvT-w24	93.23	86.06	88.93	87.03
Swin-B	92.76	85.49	90.35	87.10
Vit-B	89.34	84.61	89.82	87.83
ResNet101	85.64	78.29	84.62	83.88
Densenet201	89.82	83.62	84.77	85.00
Backbone	PBC		Thyroid	
	AUC_D	AUC_C	AUC_D	AUC_C
CvT-w24	100.00	98.26	95.04	96.49
Swin-B	100.00	98.40	95.99	96.88
Vit-B	100.00	98.31	95.53	96.84
ResNet101	99.99	98.12	92.36	94.83
Densenet201	99.99	98.03	92.93	94.57

4.6 Ablation Study

Component Ablation. As shown in Table 3, we investigate the impact of each component in our framework. When either the CRRB or the CDCI module is removed, we observe a noticeable decline in the model’s ability to capture complex concept relationships and to effectively align them with the diagnostic task. The final configuration in Table 3 demonstrates that integrating both the CRRB and CDCI modules within CREM yields optimal diagnostic performance.

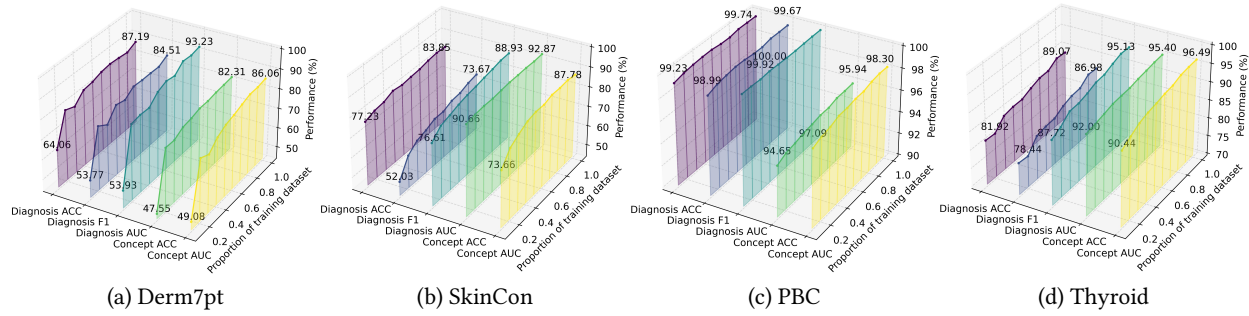


Figure 7: Disease diagnosis and concept detection performance using different portions of training data.

Backbone Ablation. As illustrated in Table 4, our method exhibits strong performance across a range of CNN- and Transformer-based backbones. Overall, the CvT-w24 and Swin-B backbones achieve the best overall performance.

4.7 Online Web Application Demo

To facilitate clinical translation, we developed DiagExplain, a web-based demo for accessible and explainable clinical decision-making. Users can upload medical images along with basic patient meta-data (e.g., age and gender) to obtain instant diagnostic predictions. Based on model predictions and concept activations, DiagExplain identifies lesion regions, provides concept-level explanations, and uses a large language model to generate structured diagnostic reports. Figures 8 and 9 in the Appendix illustrate representative cases processed by CREM. DiagExplain further incorporates a human-in-the-loop mechanism that allows clinicians to adjust concept confidence scores, refining model outputs to better align with clinical observations and enhancing the practical utility of CREM in real-world settings.

5 Conclusion

In this paper, we propose CREM, a coarse-to-fine concept-based framework for explainable medical diagnosis that overcomes the limitations of current concept-based models in relational reasoning for disease diagnosis. Unlike previous concept-based approaches that solely identify the presence of concepts to form diagnostic predictions, CREM performs final diagnostic causal inference based on holistic interactions among concepts learned through graph reasoning. Extensive experiments conducted on four public datasets demonstrate that CREM achieves state-of-the-art performance across the board. Furthermore, we deploy CREM on a web-based demo to bridge the gap between research and clinical application, enabling interpretable diagnosis.

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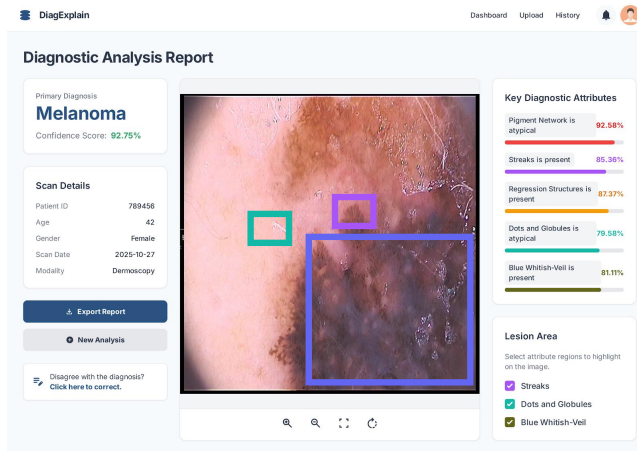
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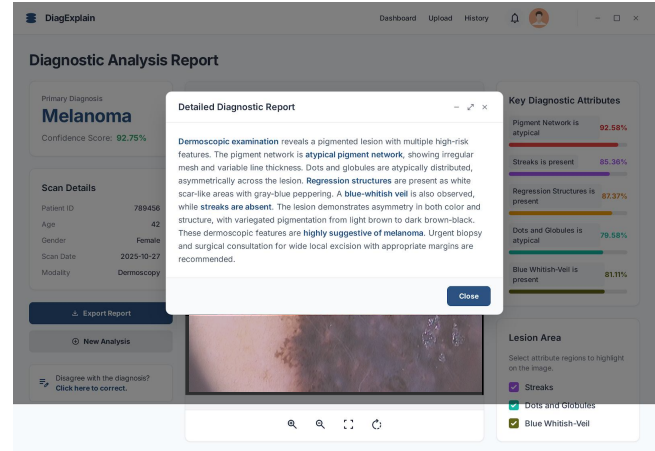
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Table 5: The dataset details in our experiments.

Dataset	Diagnosis Labels	Concept Labels
Derm7pt	Nevus (Nev), Melanoma (Mel).	Pigment Network (Absent/Typical/Atypical), Streaks (Absent/Regular/Irregular), Pigmentation (Absent/Regular/Irregular), Regression Structures (Absent/Present), Dots and Globules (Absent/Regular/Irregular), Blue-Whitish Veil (Absent/Present), Vascular Structures (Absent/Regular/Irregular).
SkinCon	Non-neoplastic (NNP), Benign (BEN), Malignant (MAL).	Papule (Absent/Present), Plaque (Absent/Present), Pustule (Absent/Present), Bulla (Absent/Present), Patch (Absent/Present), Nodule (Absent/Present), Ulcer (Absent/Present), Crust (Absent/Present), Erosion (Absent/Present), Atrophy (Absent/Present), Exudate (Absent/Present), Telangiectasia (Absent/Present), Scale (Absent/Present), Scar (Absent/Present), Friability (Absent/Present), Dome-Shaped Elevation (Absent/Present), Brown Hyperpigmentation (Absent/Present), White Hypopigmentation (Absent/Present), Purple (Absent/Present), Yellow (Absent/Present), Black Pigmentation (Absent/Present), Erythema (Absent/Present).
PBC	Basophil (BAS), Monocyte (MON), Lymphocyte (LYM), Neutrophil (NEU), Eosinophil (EOS).	Cell Size (Small/Big), Cell Shape (Round/Irregular), Nucleus Shape (six categories), Nuclear-Cytoplasmic Ratio (High/Low), Chromatin Density (Dense/Loose), Cytoplasmic Vacuoles (Absent/Present), Cytoplasmic Texture (Clear/Frosted), Cytoplasmic Colour (Purple/Blue/Light blue), Granule Type (Nil/Small/Round/Coarse), Granule Colour (Purple/Pink/Red/Nil), Granularity (Present/Absent).
Thyroid	Benign (Ben), Malignant (Mal).	Composition (Cystic/Spongiform/Mixed/Solid), Echogenicity (Anechoic/Hyperechoic-Isoechoic/Hypoechoic/Very hypoechoic), Shape (Wider-than-tall/Taller-than-wide), Margin (Smooth/Ill-defined/Lobulated/Irregular/Extra-thyroidal extension), Macrocalcifications (Without/Present), Peripheral Calcifications (Without/Present), Punctate Echogenic Foci (Without/Present).



(a)



(b)

Figure 8: Web application demos of the CREM (part 1) illustrating model decision-making: (a) primary diagnostic view with predicted diagnosis, confidence score, lesion features, and attribute probabilities; (b) detailed report providing interpretable diagnosis and clinical recommendations.

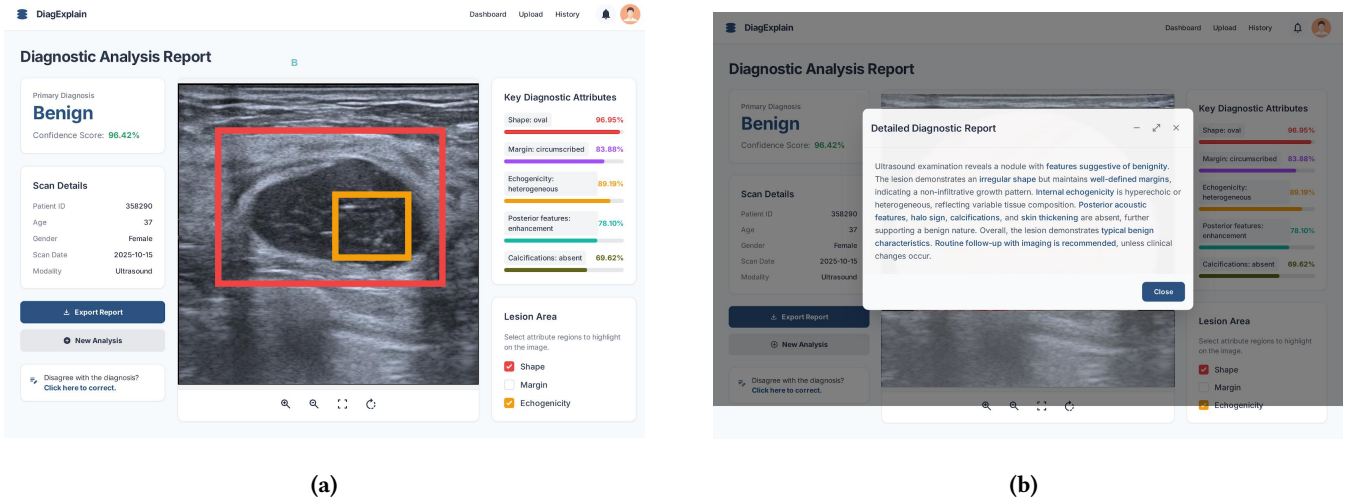


Figure 9: Web application demos of the CREM (part 2).

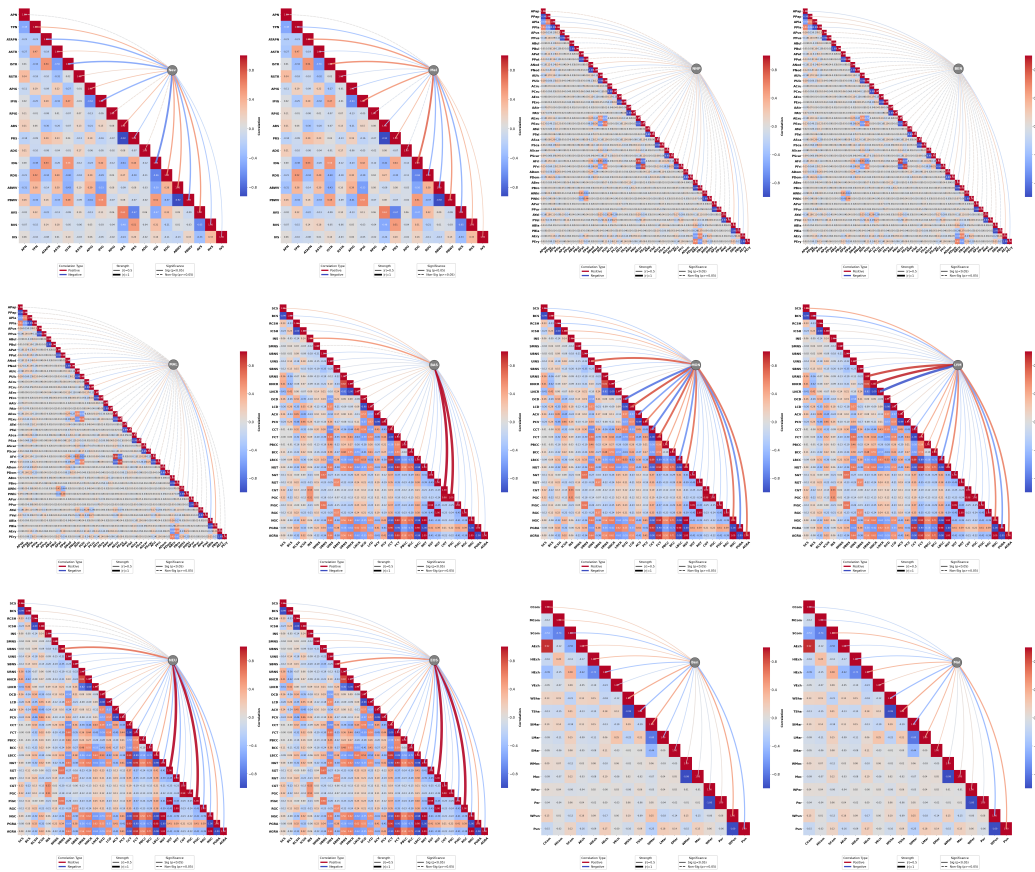


Figure 10: Each correlation network contains a lower-triangular heatmap that visualizes pairwise Pearson correlation coefficients between concept distributions, encoded through both color intensity and numeric values. The color intensity indicates the strength of inter-concept correlations, while the thickness and line style of the arcs denote the magnitude and statistical significance of each concept's contribution to the diagnostic labels. The learned concept importance shows strong alignment with clinically recognized relevance.